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Sarcasm Detection for Sentiment Analysis using Deep Learning Enhancement

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Abstract

Sarcasm is a verbal irony where the intended message contrasts with the literal meaning, often used to mock or express contempt. Sarcasm is context-dependent and requires nuanced verbal clues, tone, and context, which are challenging to capture in text-based analysis. Traditional sentiment analysis models are insufficient due to the complexity of sarcasm and cultural and individual disparities. However, new developments in deep learning, particularly transformer architecture, show promise for improving sarcasm detection accuracy. In this paper, We propose a novel method combining transformer augmentation with hybrid optimized deep learning techniques such as LSTM, BiLSTM, CNN, and pre-trained transformer-based BERT to detect complex sarcasm from social platforms. Experiments conducted on benchmark datasets reveal that our system works well, with notable gains in sarcasm detection accuracy over baseline techniques. We achieved 96% accuracy, 97% precision, 93% recall, and 95% F1-Score by the proposed model. Our model introduces a reliable and accurate way to find sarcasm in different text types by combining transformer upgrades and optimized hybrid DL techniques to benefit social media monitoring, customer feedback analysis, and opinion mining, where understanding genuine emotion is crucial.

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Keywords: Sentiment Analysis, Sarcasm Detection, Hybrid deep learning, Online Social Media, BERT;

1. Introduction

Sarcasm is linguistic irony in which the intended meaning of a remark is the reverse of its literal meaning. It is often used to ridicule or express scorn. Detecting is tricky and challenging, particularly in text, because of its tone, context, and unique language signals. Sarcasm substantially influences communication by modifying the perceived mood and purpose of remarks. According to Merriam-Webster dictionary, "The use of words that mean the opposite of what you really want to say especially in order to insult someone, to show irritation, or to be funny" [8]. Failure

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Table 1: Abbreviations

Abbreviation	Description
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
MM	Multimodal
OSN	Online Social Network
LSTM	Long short-term memory
BiLSTM	Bidirectional long short-term memory

to effectively detect sarcasm in sentiment analysis might result in misinterpretations, distorting the analysis findings and leading to erroneous conclusions. Misleading sentiment is especially troublesome in social media monitoring, customer feedback analysis, and opinion mining, where accurately comprehending genuine emotion is paramount. The complex and subtle character of sarcasm requires sophisticated detection techniques to enable precise sentiment analysis and efficient communication.

Sarcasm is context-dependent and subtle; it presents a substantial problem in textual data's natural language processing (NLP). Despite being widely used, sarcasm is typically tricky for standard models to recognize since it depends on nuanced verbal clues, tone, and context, which may be challenging to capture in text-based analysis. The lack of verbal intonations and facial gestures further hinders identification in textual communication. Furthermore, the same word might exhibit sarcasm in one scenario and sincerity in another, necessitating a profound comprehension of the current context and past information. Moreover, cultural and individual disparities in the manifestation and comprehension of sarcasm introduce an additional level of intricacy. Because of these problems, basic sentiment analysis models are not good enough. We need more advanced methods that can carefully look at complex linguistic and contextual cues to find sarcasm.

However, new developments in deep learning, especially in the transformer architecture, which is best shown by models like BERT (Bidirectional Encoder Representations from Transformers), show promise for improving the accuracy of sarcasm detection[9]. This study offers a novel method combining transformer augmentation with a hybrid optimized model to improve sarcasm detection. We use the best features of cutting-edge transformer models and hybrid deep learning approaches, including batch normalization techniques, to create a reliable and precise sarcasm detection system. By integrating transformer-based contextual awareness and hybrid RNN variants with CNN layers, our methodology seeks to overcome the drawbacks of current techniques and get better results in terms of 96% accuracy, 97% precision, 93% recall, and 95% F1-Score. This will improve accuracy in detecting sarcasm in a wide range of text sources, especially on social media sites like Twitter and Facebook, where it is common.

This research is an attempt toward a new direction for classifying unseen and uncontrolled sarcasm content on social media platforms. The proposed model is more effective than conventional techniques, with higher accuracy and lower computation resources. The rest of the paper is organized as follows: Section 2 summarizes related work for detection techniques. Table 1 includes the description of the main abbreviations used in this paper. Section 3 followed by datasets and media processing. Section 4 deals with the proposed architecture and model with generalized ability. The results and analysis are discussed in Section 5. Finally, the conclusion and future direction are in Section 6.

2. Related Work

As social media platforms continue to act as significant conduits for communication and information dissemination, the prominence of sarcasm within user-generated material offers issues for reliable statement analysis and opinion mining. Failure to identify sarcasm may lead to misunderstandings of signals, resulting in erroneous insights and incorrect decision-making processes [3]. It is essential to understand the underlying attitudes and intents of users. Hence, accuracy and effectiveness can be boosted through NLP or DL techniques. Many researchers have used different ML techniques on emotional analyses to recognize sarcastic phrases. However, it falls short of capturing complex grammatical ambiguities and contextual clues.

Using rich contextual information and semantic representations in large-scale text corpora has been made possible using DL techniques, especially BERT [15]. Using the Word2Vec, GloVe, and BERT models, they achieved a final classification accuracy of 90%, 85%, and 86% by applying fuzzy logic as ensemble embedding techniques. BERT is well-suited for tasks requiring comprehension of sarcasm and irony since it has been pre-trained on vast volumes of text data and can successfully capture the subtle nuances of language and context [4]. According to Savini and Caragea [13], recent research has shown how effective it is to fine-tune BERT for sarcasm detection on benchmark datasets. Even with the success of transformer models like BERT, sarcasm detection is still tricky, especially in loud, unstructured online spaces like Twitter [16].

To overcome these obstacles, scientists have put forward ensemble learning strategies that enhance prediction accuracy and generalization by combining many models [5]. Different authors have even used various techniques to detect sarcasm and irony. Palaniammal and Anandababu [11] have proposed a reliable method for detecting sarcasm on social media introduced utilizing the Artificial Rabbits Optimizer with Multilayer Convolutional Encoder-Decoder Neural Network (ARO-MCEDNN) methodology. Hao et al. [6] used profiling and forum topic information to detect contextual sarcasm, which enhanced semantic representations and user-forum fusion network and context-aware attention to obtain diverse representations from different aspects. He et al. [7] introduced an enhanced gate module as a text feature fusion module that decides how much information should be taken into account, as well as an adaptive incongruity extraction module to calculate the distance between each word in the text and commonsense knowledge.

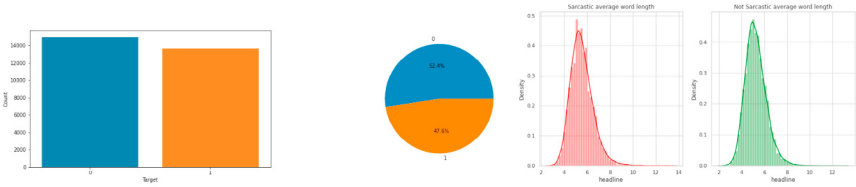
Similarly, other authors have aimed to improve sarcasm detection as s Qiao et al. [12] have proposed a Mutual-enhanced Incongruity Learning Network (MILNet) developed for multi-modal sarcasm detection. A global learning module and a local semantic-guided learning module were added to improve performance, allowing the two modules to benefit from their consistency. Tomás et al. [17] suggested a framework for combining text and images that use visual transformers to help separate text attention. This improved the F1-score by as much as 9% compared to the best existing visio-linguistic transformers and earlier research in the field. Ali et al. [2] suggest a deep learning model that uses a GlobalMaxPool1D layer to get better at extracting features and LSTM to figure out how news headlines relate to each other in terms of time. Alaramma et al. [1] used SVM classifiers combined with lexical, pragmatic, linguistic incongruity, and context insecurity characteristics have been suggested as part of an ensemble classification technique for sarcasm detection. Sharma et al. [14] proposed a novel hybrid sentence embedding-based technique with an autoencoder on three real-world social media datasets: Twitter, headlines, and the Self-Annotated Reddit Corpus.

3. Datasets and Media Processing

Picking the suitable dataset and getting it ready are essential steps to ensure that a hybrid deep learning model works well and can be used in other situations. Various textual and multimodal sarcasm-related datasets are available on social media sites. However, previous researchers have used most of the Twitter or other social media datasets gathered using hashtag-based supervision. These datasets include a lot of linguistic and label noise. Moreover, many tweets respond to others and need contextual tweets to identify sarcasm. Hence, in this proposed work, the "*News Headlines dataset for Sarcasm Detection*" has been used, taken from Kaggle, and used in different research articles [10]. This dataset was gathered from two news websites to overcome the noise-related restrictions on Twitter datasets. To provide satirical takes on current affairs, from all sarcastic headlines from the News in Breid and News in Photo categories, further actual news headlines from HuffPost that aren't ironic. It contains 28,619 records, of which 13,635 are sarcastic, and 14,984 are non-sarcastic records. The reason for selecting this dataset was that there are no typos or casual language in news headlines since they are composed by experts in a formal style. In addition to decreasing sparsity, this raises the likelihood of discovering pre-trained embeddings.

3.1. Data Analysis

Data exploration, cleansing, visualization, and modeling are essential phases in data analysis. Data cleaning entails locating and fixing mistakes, missing values, and inconsistencies, while exploration compiles and displays data using graphs, chats, and descriptive statistics to find patterns or trends. Here, we plotted the data distribution to inspect whether the dataset is imbalanced or balanced 1a. Observing the 1a, it indicates that the dataset is almost balanced, and there is no need for any further approach to imbalanced data. Figure 1b represents the average world length



(a) Data Distribution (b) Average Word Length

Fig. 1: Data Visualization and Pre-processing



(a) Sarcastic Words (b) Not Sarcastic Words

Fig. 2: Most Sarcastic and Nor Sarcastic Words

of texts that helps to build a model precisely by better understanding the vocabulary richness, syntactic complexity, and semantic diversity of the text corpus. Tokenization techniques, embedding dimensionality selection, and input sequence length determination are just a few building model and optimization components that may benefit from the average word length.

Initially, stop words and special characters were eliminated from textual datasets. Additionally, statistically examined the text's substance. Here *Word Cloud* 2 used for this. A word cloud is a textual data visualization in which phrases provide a rapid and easy-to-understand summary of the main ideas, themes, and subjects often encountered in the text by displaying worlds in different sizes according to how frequently they appear. While preparing data, it guides the elimination of noisy words. Several *noise words* often occur in the dataset but don't contribute to comprehension or semantic value. Words without any meaning or information, such as *the*, *a*, *an*, and so on, are considered noisy words since they often occur in text data. Before doing more analysis, we created a list of stop words to exclude noisy terms from the text data.

3.2. Media Preprocessing

Preprocessing processes are necessary to standardize and prepare the textual input for analysis before training the model. By providing useful contextual information that improves the precision and potency of the detection algorithm, media processing is essential to the detection of sarcasm. Tokenization, lowercase, elimination of stop words and special characters, and stemming or lemmatization to standardize some activities have been used before sending to the model. Images or videos might provide extra contextual clues important for sarcastic identification, Still it must be extracted from the dataset because it needs extra computer vision techniques with a colossal computation cost. The proposed model includes contextual information using text embedding or feature extraction from emojis and other textual contexts.

Features associated with text alone are insufficient to identify all types of sarcasm that may exist. DL techniques directly fail to use all the constituents available in a tweet. Hence, other characteristics need to be retrieved. Sarcasm is a complex kind of discourse that manipulates words and meanings by using behavioral elements such as # Total count of exclamation marks, total count of question marks, total count of dots, total count of all-capital terms, total count of quotations. Intonation and facial expressions can be used if audio and video contents are available in dataset. Here, text carefully examined while preprocessing that any of the words have a vowel which is repeated more than

twice (e.g., ”looooove”). Should a word of such kind exist, the feature is assigned a true value. Otherwise, the value is stored as false. Also, the disproportionate use of exclamation marks or question marks, especially in phrases that evoke strong emotions, may indicate a certain tone the user wants to convey. Nevertheless, this tone is not always caustic.

After preprocessing, the dataset is split into a 7:3 ratio for training and validation, and the model uses selected parameters: the learning rate is 0.001, stochastic gradient descent (SGD) for optimization, and binary cross-entropy as loss function. The same optimized parameters have been used for different RNN models used for feature learning. Different batch sizes have been tried with different numbers of epochs up to 100, but after 20 epochs and 64 batch sizes, having almost optimal performance.

4. The Proposed DL Architecture

The proposed hybrid DL framework attempts to identify sarcasm in online social content correctly. The suggested structure comprises sub-models that do clear things like data collection, pre-processing, feature engineering, and classification, as shown in Figure 3. It combines traditional ML techniques with more advanced DL structures, like BiLSTm and BERT with CNN layers. Furthermore, the model incorporates contextual comprehension processes, such as attention mechanisms or bidirectional processing, to accurately represent the contextual subtitles of sarcasm. Here, different variants of RNNs and BERT have been used for feature extraction then sent to the CNN layers and finally goes to Softmax for classification. After repeated training and refining, the suggested structure differentiated between sarcastic and non-sarcastic.



Fig. 3: The Proposed Hybrid Deep Learning Architecture

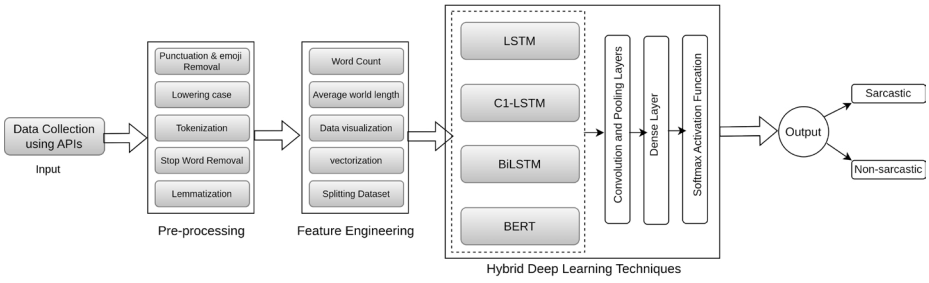
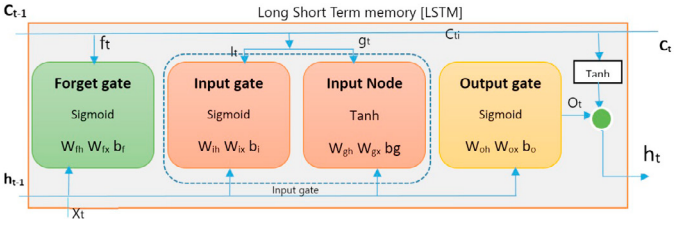


Fig. 4: Generalized Deep Learning Model for Sarcasm Detection

The Generalized Hybrid Deep Learning Model for Sarcasm Identification is a flexible framework that integrates the advantages of several deep learning approaches to detect sarcastic expressions in text-based data accurately. This model incorporates many components, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer-based models like BERT, to effectively capture various facets of sarcasm. The proposed model consists of different sub-modules that have well-defined functionality. Some preprocessing steps eliminate the unwanted distortions, as shown in the pre-processing sub-module in Figure 4, then perform tokenizing on input text. The input text can be extracted using API provided by social media platforms. The feature engineering sub-module is responsible for feature creation, transformation, data visualization, and selection. After splitting the dataset into training and validation, these traits are sent through advanced hybrid deep learning architectures to pick up on broad connections and temporal context, which are necessary to understand sarcasm. We have used transformer-based architectures to take advantage of contextual embeddings and fully capture dependencies across the whole input sequence. The proposed architecture combines standard deep-learning techniques for feature extraction and, finally Softmax Activation function to differentiate sarcasm accurately and reliably.

4.1. Hybrid Sub-Model Components

A type of recurrent neural network (RNN) architecture called Long Short-Term Memory (LSTM) can find long-range dependencies in sequential data by fixing the problem of vanishing gradients as shown in Figure 5. It uses hyperbolic tangent (tanh) and sigmoid activation functions, and it has different gating mechanisms that control the flow of information through the network. This lets the network selectively send information. The forget gate (f_t) chooses which data from the previous cell state should be deleted, while the input gate (i_t) selects which data from the current input should be stored in the memory cell. The output gate ((o_t)) decides what information from the current cell state should go to the following layer or timestep. A candidate memory cell ($\tilde{c}_t$) and an update mechanism to change the cell state (c_t) are also included in LSTM. The output gate and the hyperbolic tangent of the updated cell state are used to calculate the hidden state (h_t), which is the LSTM cell's output. Matrix multiplications, element-wise operations, and activation functions are the driving forces behind LSTM processes. It provides a solid base for modeling sequential data with long-term dependencies.



(a) Schematic diagram of LSTM

$$
\begin{aligned}
f_t &= \sigma[(W_{fh} \cdot h_{t-1}) + (w_{fx}.x_t) + b_f] \\
i_t &= \sigma[(W_{ih} \cdot h_{t-1}) + (w_{ix}.x_t) + b_i] \\
\tilde{g}_t &= tanh[(W_{gh} \cdot h_{t-1}) + (w_{gx}.x_t) + b_g] \\
C_t &= f_t * C_{t-1} + i_t * \tilde{g}_t \\
o_t &= \sigma[(W_{oh} \cdot h_{t-1}) + (w_{ox}.x_t) + b_o] \\
h_t &= ot \odot tanh(C_t)
\end{aligned}
$$

(b) LSTM Equations

Fig. 5: LSTM schematic diagram and its equations

BiLSTM is an extension of LSTM architecture that uses bidirectional processing to store sequential data containing past and future information. This bidirectional processing improves the model's comprehension and representation of sequential input by capturing relationships from past and future contexts. BiLSTM is composed mathematically of two LSTM layers, one of which processes the input sequence forward ($\overrightarrow{LSTM}$) and the other backward ($\overleftarrow{LSTM}$). The final output representation (h_t) is formed by concatenating the hidden states ($\overrightarrow{h_t}$ and $\overleftarrow{h_t}$) that are calculated by each LSTM layer at time step t. With different values for each direction, the equations controlling the forward and backward LSTM operations are comparable to those of the regular LSTM. To be more precise, the forward LSTM computes the hidden state $\overrightarrow{h_t}$ using the following information: input sequence x_t, previous hidden state $\overrightarrow{h_{t-1}}$, and cell state $\overrightarrow{c_{t-1}}$. On the other hand, the backward LSTM computes $\overleftarrow{h_t}$ using the reverse input sequence, previous hidden state $\overleftarrow{h_{t+1}}$, and cell state $\overleftarrow{c_{t+1}}$. Concatenating $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ yields the final output h_t.

A hybrid convolutional LSTM model takes the best parts of both convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) and uses them together to handle and understand time- and space-dependent sequential input well. In this architectural design, CNN layers extract features from input sequences, effectively capturing spatial patterns and local interdependencies. On the other hand, LSTM layers are applied to represent temporal dependencies and effectively capture extensive contextual information. The CNN layers capture salient characteristics from the input data, which are then input into the LSTM layers for sequential modeling and prediction.

Bidirectional Encoder Representations from Transformers (BERT) is an innovative paradigm based on transformers that utilizes a multi-layer bidirectional transformer architecture to extract contextual information from previous and following words in a sentence. Feedforward neural networks and self-attention processes are present in every transformer layer. The self-attention mechanism calculates attention ratings for each pair of words in a phrase to identify dependencies and contextual links. Next, while creating representations, the attention ratings are used to determine how important each word is. After that, a series of feedforward neural networks work on these representations to give each word in the input sequence a contextualized embedding.

5. Implementation and Results Analysis

The experimental findings demonstrate how well the suggested design generalizes across social media sites with distinct datasets. The results show that the proposed architecture successfully detects sarcasm, achieving high levels of recall, accuracy, precision, and F1-score on the validation set. We used the latest Python package on a local GPU-based workstation with 4 GB of RAM to realize our idea. The *TensorFlow* framework uses the Keras APIs and the Python libraries Sklearn, Pandas, Numpy, Matplotlib, etc. for data analysis. To adjust each neural network's ability to generalize, we used a variety of neural networks (NNs) and their variations with adjusted hyperparameters.

5.1. Performance of the Proposed Model

The performances of the proposed model mentioned above are shown in Table 2 for both sarcastic and non-sarcastic contents concerning accuracy, precision, recall, and F1-Score. The table's bolded text displays the optimal results from various combinations of deep learning models. With the combination BERT, the last rows of the table include the majority of the top performance metrics.

Started our experiments by using LSTM with CNN layers to learn textual or contextual representation. LSTM was selected because of its capacity to learn and retain pertinent contextual information over long periods. It also handles different data lengths efficiently without vanishing gradient issues. The first row of Table 2 shows the performance of the LSTM model using that we obtained 84% accuracy. We achieved 85% precision, 84% recall, 85% F1-Score for sarcastic whereas 85% precision, 83% Recall, 82% F1-score for Non-Sarcastic.

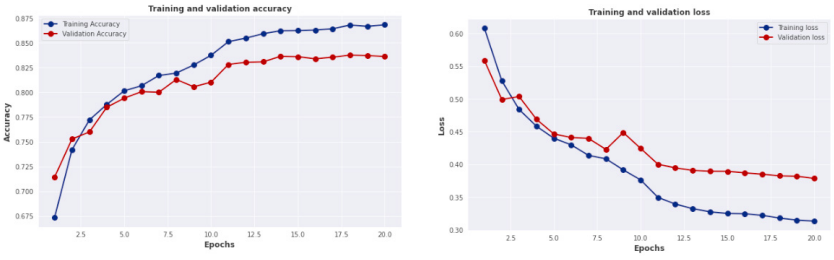
Table 2: Classification Results of the proposed model

Method	Accuracy	Sarcastic			Not Sarcastic		
		Prec	Recall	F1 score	Prec	Recall	F1
LSTM	0.84	0.85	0.84	0.85	0.82	0.83	0.82
C1-LSTM	0.85	0.84	0.84	0.84	0.85	0.85	0.85
BiLSTM	0.86	0.85	0.85	0.85	0.87	0.87	0.87
BERT	0.96	0.97	0.93	0.95	0.94	0.97	0.96

With the advantage of LSTM, there are some limitations, such as poor generalization capability [18]. Hence, a convolutional layer was applied in the proposed architecture before LSTM and kept the remaining architecture the same; using this hybridized architecture, the accuracy improved from 84% to 85%, which is better than the previous, but there was only improvement in Non-Sarcastic content. Hence, an extension version of LSTM, BiLSTM (Bidirectional Long Short-Term Memory) is used. Here, overall accuracy improved 86% and 85% recall for sarcastic content and also showed some improvement for non-sarcastic content. Comparing to the previous performance gain by LSTM for sarcastic content, it has less improvement as compare to non-sarcastic content. Hence, we decided to use pre-trained advanced deep-learning BERT models, and got an overall better performance comparing the previous model as 96% accuracy.

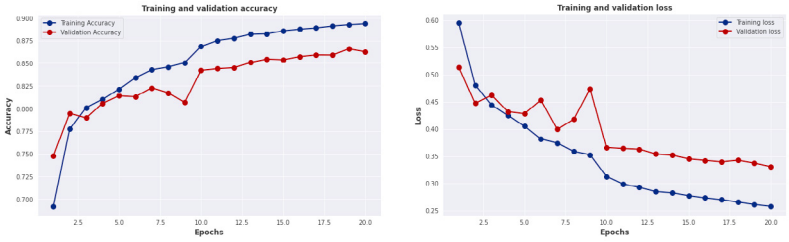
5.2. Training and Validation Datasets: Loss and Accuracy

A modified hybrid DL model is used for shallow and contextual information to improve overall sarcasm detection skills. This proposed hybrid model improves accuracy and robustness by combining BERT's sophisticated linguistic subtleties understanding with the capabilities of older algorithms in capturing surface-level patterns. The overall performance is 96% accuracy, which is optimal accuracy and quite better than some of SoTA methods. The precision, Recall and F1-score for sarcastic and non-sarcastic content are improved and reached almost optimal compared to previous models, which can be visualized from Table 2.



(a) Accuracy (b) Loss

Fig. 6: Training and Validation Accuracy and Loss using LSTM



(a) Accuracy (b) Loss

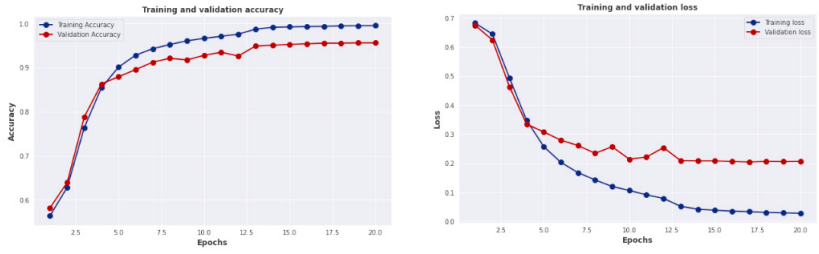
Fig. 7: Training and Validation Accuracy and Loss on BiLSTM

Training and validation curves provide succinct visual depictions of performance regarding the functionality of proposed models and the capacity for generalization. These curves (Figure 6) represent the accuracy and loss of training and validation data over successive epochs and how effectively the model converges towards an optimal solution and aids in identifying overfitting or underfitting. The accuracy curve demonstrates the model's capacity to accurately classify instances in the training dataset across consecutive epochs in an upward trend as the model improves its ability. In contrast, the loss curve illustrates the model's inaccuracy (errors), where smaller numbers indicate superior performance. A diminishing loss curve indicates that the model is approaching an ideal solution, but oscillation or stagnation may reflect difficulties in convergence or poor training settings. By examining these curves, researchers may track the learning progress of the model, detect any problems like overfitting or underfitting and make informed modifications to the model's structure, hyperparameters, or training methods to enhance performance and reach the target accuracy levels.

Figure 6 depicts the accuracy and loss curves for training and validation data using LSTM techniques. We trained and optimized our model with different hyperparameters, such as learning rate, dropout, batch size, etc. Figure 6a shows the accuracy of training and validation data, where training starts at 67% and goes up to 87%, whereas validation accuracy starts at 71% and goes up to 84% after 20 epochs. We tried with higher numbers of epochs, but after 20 epochs, performance was almost stabilized. In contrast, Figure 6b indicates that the training and validation losses should decrease over time. Here, training loss started at around 0.60 and reached 0.0. Similarly, validation loss dropped from 0.55 to 0.38.

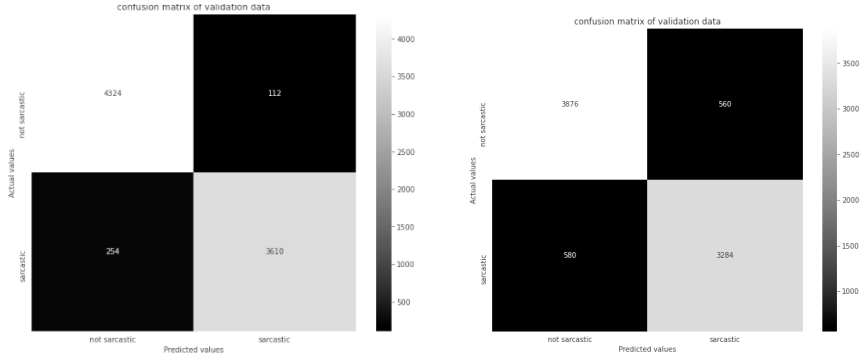
Figure 7 depicts the performance proposed model using BiLSTM on training and validation data. Training accuracy (Figure 7a) starts from 70% and goes up to 89%, whereas validation accuracy starts at 75% and goes up to 85%. Similarly, training loss (Figure 7b) began at 0.60 and came down to 0.25, and validation loss started at 0.51 and reached 0.34. In the same way, 8 depicts the performance of the proposed model using BERT. Here, training and validation accuracy both started at 55% and went up to 99% and 96%, respectively. Figure 8b indicates the training and validation loss, where both started at 0.7 and came down to 0.02 and 0.2, respectively.

Confusion matrices (CM) provide a concise and enlightening summary of a model's classification. Each row of the matrix corresponds to the examples in a predicted class, whereas each column corresponds to the instances in an



(a) Accuracy (b) Loss

Fig. 8: Training and Validation Accuracy and Loss using BERT



(a) BERT (b) LSTM

Fig. 9: Confusion Matrics

actual class. The diagonal members of the matrix correspond to be appropriately categorized, while the off-diagonal elements reflect occurrences that have been misclassified. The confusion matrix provides four key measures: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). These metrics calculate accuracy, precision, recall, and F1-score assessment metrics. Figure 9 depicts the CM of different techniques used on the same dataset with a total validation sample size of 8,300 after preprocessing and splitting into training and validation data. The validation dataset contained 4,436 non-sarcastic data, and 3,864 were sarcastic data.

By evaluating CM 9a the total actual non-sarcastic value 4,436 (4324 + 112) was considered for classification, and the model predicted exactly 4,578(4324 + 254) non-sarcastic values. Similarly, 3,864(254 + 3610) actual sarcastic values were considered for validation, out of which the model correctly predicts sarcastic 3,722(3610 + 112)values. As a result, 3610 TP, 4324 FN, 254 FN, and 112 FP values.

6. Conclusion

Sarcasm is widely considered the holy grail for social media platforms and microblogging websites, which are tricky and challenging to detect. In the proposed work, a significant improvement has been achieved for automated sarcasm detection systems using hybrid deep learning models. It includes transformer-based architecture BERT and convolutional neural network layers with better generalization ability using normalization techniques. By exploiting contextual understanding, the proposed method shows better performance and robustness in the sarcastic identification of textual data. Here, bidirectional encoder representations from transformers and other DL models, such as LSTM and BiLSTM, have been used for feature learning. Different statistical and linguistic features have been considered and integrated into contextual learning. The experiment results demonstrate the efficacy of an optimized hybrid model through testing and validation, where accuracy and other performance matrices indicate its potential use in a real-

world dataset with an overall accuracy of 96%. Our study adds to the continuing efforts to improve natural language processing methods for sarcasm detection.

Data and Code Availability. All data used in this work is publicly available. The program code and artifacts, through the *GitHub* repository[1], necessary to run and execute for interpreting, replicating, and building on the findings reported in the paper, will be made publicly available on publication of this paper.

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